While the referral did not actually identify a divergence in the case law, the Enlarged Board of Appeal considered that there was at least the potential for confusion, arising from the assumption that **any technical considerations** were sufficient to confer technical character on claimed subject-matter, a position which was apparently adopted in some cases (e.g. **T 769/92**). **T 769/92** (OJ 1995, 525) was an example of an invention which concerned the internal functioning of a computer caused by the programs running on it. According to this decision the fact that technical considerations were required in order to arrive at the invention was considered to lend sufficient technical character to the invention as claimed for it to avoid exclusion from patentability under Art. 52(2)(c) and (3) EPC 1973, whereas no importance was attributed to the specific use of the system as a whole.

However, **T 1173/97** set the barrier higher in the case of computer programs. It argued that all computer programs have **technical effects**, since for example when different programs are executed, they cause different electrical currents to circulate in the computer they run on. However, such technical effects are **not sufficient to** confer "technical character" on the programs; they must cause **further technical effects**. In the same way, it seemed to this board, that although it may be said that all computer programming involves technical considerations since it is concerned with defining a method which can be carried out by a machine, that in itself is not enough to demonstrate that the program which results from the programming has technical character; the programmer must have had technical considerations beyond "merely" finding a computer algorithm to carry out some procedure.

The board in **T 1173/97** concentrated on the effect of carrying out an algorithm on a computer, noting that there were always technical effects, which led the board, since it recognised the position held by the framers of the EPC, to formulate its requirement for a "further" technical effect. Only if a computer program, when run, produced **further technical effects**, was the program to be considered to have a technical character. In the same way, it would appear that the fact that fundamentally the formulation of every computer program requires technical considerations in the sense that the programmer has to construct a procedure that a machine can carry out, is not enough to guarantee that the program has a technical character (or that it constitutes "technical means" as that expression is used in e.g. **T 258/03**). By analogy one would say that this is only guaranteed if writing the program requires "further technical considerations".

In **T 598/14** the application related to a method for generating, from an input set of documents, a word replaceability matrix defining semantic similarity between words occurring in the input document set. The board considered that the translation, with the aim of enabling the linguistic analysis to be done automatically by a computer, could be seen as involving, at least implicitly, technical considerations. This was also in line with decision **T 1177/97** or opinion **G 3/08** (OJ 2011, 10). However, according to **G 3/08**, point 13.5 of the Reasons, this is not enough to guarantee the technical character of subject-matter otherwise excluded from patentability under Art. 52(2) and (3) EPC. The technical character would have to be established on the basis that those considerations constituted "further technical considerations". The board concluded that the subject-matter of independent claim 1 lacked an inventive step (Art. 52(1) EPC and 56 EPC).

In T 2825/19, the board considered that opinion G 3/08, when compared to decision T 1173/97, reframed the interpretation of technicality with respect to computer programs. G 3/08 (point 13.5 and 13.5.1 of the Reasons), explicitly rejected the position that any technical considerations are sufficient to confer technical character on claimed subject-matter. Such a narrower interpretation of the term "technical" with respect to computer programs is a normal development for the interpretation of a legal provision open to interpretation (see opinion G 3/19), and this was the case for "programs for computers" "as such" in Art. 52(2)(c) and (3) EPC. The board understood opinion G 3/08 as taking a negative view on the technical character of the activity of programming a computer. Computer hardware is without any doubt a field of technology within the meaning of Art. 52(1) EPC. Consequently, the board saw no reason why considerations that specifically exploit technical properties of the computer system hardware to solve a technical problem related to the internal operation of the computer system, such as storing data in main memory instead of on a hard disk to be able to read the stored data with less delay, should not be viewed as "further technical considerations" in accordance with opinion G 3/08. Such considerations (and associated "further" technical effects) were not present in all computer programs. By contrast, the board saw no support for the appellant's view that the concept "further technical considerations" should be interpreted with a broader meaning that would also cover considerations aiming to solve problems "merely" relating to programming.

g) Technical considerations: implementation of a function on a computer system

In T 1177/97 claim 1 was directed to a method for translation between natural languages; accordingly, it used various linguistic terms and involved linguistic aspects of the translation process. The board raised the question whether such linguistic concepts and methods could form part of a technical invention at all. It referred to EPO case law which provided various examples showing that even the automation of such methods did not make good a lack of technical character (e.g. T 52/85). On the other hand, coded information had been considered, on a case-by-case basis, as a patentable entity, i.a. T 163/85, OJ 1990, 379; T 769/92, OJ 1995, 525 and T 1194/97, OJ 2000, 525. The board confirmed that, in accordance with this case law, it seemed to be common ground that the use of a piece of information in a technical system, or its usability for that purpose, could confer a technical character on the information itself, in that it reflected the properties of the technical system, for instance by being specifically formatted and/or processed. When used in or processed by the technical system, such information could be part of a technical solution to a technical problem and thus form the basis for a technical contribution of the invention to the prior art.

In so far as technical character was concerned, the board stressed that it should be irrelevant that the piece of information was used or processed by a conventional computer, or any other conventional information processing apparatus, since the circumstance that such an apparatus had become a conventional article for everyday use did not deprive it of its technical character, just as a hammer still had to be regarded as a technical tool even though its use had been known for millennia. The board thus came to the conclusion that information and methods related to linguistics could in principle assume a technical character if they were used in a computer system and formed part of a technical problem solution. Implementing a function on a computer system **always involved technical considerations**, at least implicitly, and meant in substance that the functionality of a technical system was increased. The implementation of the information and methods related to linguistics as a computerised translation process similarly required technical considerations and thus provided a technical aspect to per se non-technical things such as dictionaries, word matching or the translation of compound expressions into a corresponding meaning. Features or aspects of the method which reflected only peculiarities of the field of linguistics, however, had to be ignored in assessing inventive step.

Decision **T 115/85** (OJ 1990, 30) concerned a method for displaying one of a set of predetermined messages comprising a phrase made up of a number of words, each message indicating a specific event which might occur in the input-output device of a word processing system which also included a keyboard, a display and a memory. The board observed that automatically giving visual indications of conditions prevailing in an apparatus or a system was basically a technical problem. The application proposed a solution to this technical problem involving the use of a computer program and certain tables stored in a memory. It adopted the principle laid down in decision **T 208/84**: an invention which would be patentable in accordance with conventional patentability criteria should not be excluded from protection by the mere fact that for its implementation modern technical means in the form of a computer program are used. However, it did not follow from this that conversely a computer program could under all circumstances be considered as constituting technical means. In the case in question the subject-matter of the claim, phrased in functional terms, was not barred from protection by Art. 52(2)(c) and (3) EPC 1973 (see also **T 790/92**).

h) Methods performed by a computer

In **T 258/03** (OJ 2004, 575) the board held that a method involving technical means is an invention within the meaning of Art. 52(1) EPC 1973 (as distinguished from decision **T 931/95**, OJ 2001, 441). Thus with regard to methods, decision **T 931/95** was explicitly overturned by **T 258/03**. If the claimed method requires the use of a computer, it has technical character and constituted an invention within the meaning of Art. 52(1) EPC (see **T 258/03**, **T 1351/04**, **T 313/10**). Since a claim directed to a method of operating a computer involved a computer it could not be excluded from patentability by Art. 52(2) EPC (**G 3/08**, OJ 2011, 10).

In **T 313/10** the examining division had argued, using their own criteria, that a method performed by a computer was excluded. The first issue in this case was whether the claimed method, performed by a computer, of matching items in a table is excluded from patentability (Art. 52(2) and (3) EPC). The board stated that it is the established case law of the boards of appeal that claimed subject-matter specifying **at least one feature** not falling within the ambit of Art. 52(2) EPC is not excluded from patentability by the provisions of Art. 52(2) and (3) EPC (see **G 3/08**, OJ 2011, 10; **T 258/03** and **T 424/03**).

The board noted that the technical character might come from within, namely from the effect on the computer. This was the case, for example in **T 424/03** where the technical effect came from "functional data structures (clipboard formats) used independently of any cognitive content...in order to enhance the internal operation of a computer system". Such "functional data structures" were also considered to be present in the file search method that was the subject of **T 1351/04**.

i) Computer-implemented simulation methods

G 1/19 concerned the computer-implemented simulation of the movement of a pedestrian crowd through an environment such as a building. The questions of law referred to the Enlarged Board of Appeal were answered as follows:

1. A computer-implemented simulation of a technical system or process that is claimed as such can, for the purpose of assessing inventive step, solve a technical problem by producing a technical effect going beyond the simulation's implementation on a computer.

2. For that assessment it is not a sufficient condition that the simulation is based, in whole or in part, on technical principles underlying the simulated system or process.

3. The answers to the first and second questions are no different if the computer-implemented simulation is claimed as part of a design process, in particular for verifying a design.

In the Enlarged Board's opinion, the COMVIK approach was suitable for the assessment of computer-implemented simulations. Like any other computer-implemented inventions, numerical simulations may be patentable if an inventive step can be based on features contributing to the technical character of the claimed simulation method. When the COMVIK approach is applied to simulations, the underlying models form boundaries, which may be technical or non-technical. In terms of the simulation itself, these boundaries are not technical. However, they may contribute to technicality if, for example, they are a reason for adapting the computer or its functioning, or if they form the basis for a further technical use of the outcomes of the simulation (e.g. a use having an impact on physical reality). In order to avoid patent protection being granted to non-patentable subject-matter, such further use has to be at least implicitly specified in the claim. The same applies to any adaptations of the computer or its functioning. The same considerations apply to simulations claimed as part of a design process.

Whether a simulation contributes to the technical character of the claimed subject-matter does not depend on the quality of the underlying model or the degree to which the simulation represents "reality". However, the accuracy of a simulation is a factor that may have an influence on a technical effect going beyond the simulation's implementation and may therefore be taken into consideration in the assessment under Art. 56 EPC (see also **T 489/14** of 26.11.2021).

A simulation is necessarily based on the principles underlying the simulated system or process. Even if these principles can be described as technical, the simulation does not necessarily have a technical character. The Enlarged Board was of the opinion that it is neither a sufficient nor a necessary condition that a numerical simulation is based, at least in part, on technical principles that underlie the simulated system or process.

In coming to its decision, the Enlarged Board also looked at the decision in **T 1227/05** (OJ 2007, 574), which had dealt with an application relating to a computer-implemented method with mathematical steps for simulating the performance of a circuit subject to 1/f noise. The solution was based on the notion that 1/f noise can be simulated by feeding suitable random numbers into the circuit model. In the deciding board's view, the simple generation of the random numbers and the possibility of calculating them separately, before the start of the circuit simulation, provided for a resource-efficient computer simulation. In its analysis under Art. 56 EPC, the board explicitly relied on the COMVIK approach, finding that the simulation of a circuit subject to 1/f noise constituted an adequately defined technical purpose for a computer-implemented invention "provided that the method is functionally limited to that technical purpose". In view of the method's functional limitation to the simulation of a noise-affected circuit, the board came to the conclusion that such simulation could be considered to be a functional technical feature. The board also made clear that the metaspecification of an (undefined) technical purpose could not be considered adequate.

In **G 1/19** the Enlarged Board, regarding **T 1227/05**, stated that calculated numerical data reflecting the physical behaviour of a system modelled in a computer usually cannot establish the technical character of an invention in accordance with the COMVIK approach, even if the calculated behaviour adequately reflects the behaviour of a real system underlying the simulation. Only in exceptional cases may such calculated effects be considered implied technical effects (for example, if the potential use of such data is limited to technical purposes). However, the Enlarged Board stated that its role was not to re-assess decision **T 1227/05**, which was taken in the specific circumstances of the case. It also noted that the board in **T 1227/05** did not rely for its decision solely on its findings that the simulated system was a technical system and that the system could only be understood and modelled by relying on technical considerations.

The Enlarged Board in **G 1/19** also referred to **T 625/11**, which concerned a method for establishing by a computer system at least one limit value for at least one operational parameter of a nuclear reactor, which method included a simulation step and resulted in numerical value(s) for one or more limit values for e.g. global power P of the reactor. The board in **T 625/11** considered that the relevant questions were the same as in **T 1227/05** and ultimately followed the conclusions of that decision, accepting that the calculated limit values for the operation of a nuclear reactor conferred a technical character to the invention. The Enlarged Board agreed with the findings of **T 1227/05** and **T 625/11** **if they were understood as being that the claimed simulation processes in those particular cases possessed an intrinsically technical function**. However, there were rather strict limits for the consideration of potential or merely calculated technical effects according to the COMVIK approach. The often-quoted criterion of **T 1227/05** that the simulation constituted an adequately defined technical purpose for a numerical simulation method if it was functionally limited to that purpose should not be taken as a generally applicable criterion of the COMVIK approach for computer-implemented simulations, since the findings of **T 1227/05** were based on specific circumstances which did not apply in general.